MOVIE RATING PREDICTION

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.ensemble import RandomForestRegressor
```

```
movies = pd.read_csv("/content/IMDb Movies India.csv",encoding='latin1')
```

```
movies
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
0		NaN	NaN	Drama	NaN	NaN	J.S. Randhawa	Manmauji	Birbal	Rajendra Bhatia
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
2	#Homecoming	(2021)	90 min	Drama, Musical	NaN	NaN	Soumyajit Majumdar	Sayani Gupta	Plabita Borthakur	Roy Angana
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
4	...And Once Again	(2010)	105 min	Drama	NaN	NaN	Amol Palekar	Rajat Kapoor	Rituparna Sengupta	Antara Mali
...	...	...	...	...	...	...	...	...	...	...
15504	Zulm Ko Jala Doonga	(1988)	NaN	Action	4.6	11	Mahendra Shah	Naseeruddin Shah	Sumeet Saigal	Suparna Anand
15505	Zulmi	(1999)	129 min	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani
15506	Zulmi Raj	(2005)	NaN	Action	NaN	NaN	Kiran Thej	Sangeeta Tiwari	NaN	NaN
15507	Zulmi Shikari	(1988)	NaN	Action	NaN	NaN	NaN	NaN	NaN	NaN
15508	Zulm-O-Sitam	(1998)	130 min	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja

15509 rows × 10 columns

Next steps: New interactive sheet

```
movies = movies.dropna(subset=['Rating'])
```

```
movies
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	(1997)	147 min	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	(2005)	142 min	Drama, Romance, War	7.4	1,086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	(2012)	82 min	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15501	Zulm Ki Hukumat	(1992)	NaN	Action, Crime, Drama	5.3	135	Bharat Rangachary	Dharmendra	Moushumi Chatterjee	Govinda
15503	Zulm Ki Zanjeer	(1989)	125 min	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15504	Zulm Ko Jala Doonga	(1988)	NaN	Action	4.6	11	Mahendra Shah	Naseeruddin Shah	Sumeet Saigal	Suparna Anand
15505	Zulmi	(1999)	129 min	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani

Next steps: New interactive sheet

```
movies['Genre'] = movies['Genre'].fillna('Unknown')
movies['Director'] = movies['Director'].fillna('Unknown')
movies['Actor 1'] = movies['Actor 1'].fillna('Unknown')
movies['Actor 2'] = movies['Actor 2'].fillna('Unknown')
movies['Actor 3'] = movies['Actor 3'].fillna('Unknown')

movies['Duration'] = movies['Duration'].fillna('0 min')
movies['Votes'] = movies['Votes'].fillna('0')
movies['Year'] = movies['Year'].fillna('(0)')
```

```
/tmp/ipykernel_3610/2189151955.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Genre'] = movies['Genre'].fillna('Unknown')
/tmp/ipykernel_3610/2189151955.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Director'] = movies['Director'].fillna('Unknown')
/tmp/ipykernel_3610/2189151955.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Actor 1'] = movies['Actor 1'].fillna('Unknown')
/tmp/ipykernel_3610/2189151955.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Actor 2'] = movies['Actor 2'].fillna('Unknown')
/tmp/ipykernel_3610/2189151955.py:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Actor 3'] = movies['Actor 3'].fillna('Unknown')
/tmp/ipykernel_3610/2189151955.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
```

```
  movies['Duration'] = movies['Duration'].fillna('0 min')
/tmp/ipykernel_3610/2189151955.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Votes'] = movies['Votes'].fillna('0')
/tmp/ipykernel_3610/2189151955.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Year'] = movies['Year'].fillna('(0)')
```

```
movies
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	(1997)	147 min	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	(2005)	142 min	Drama, Romance, War	7.4	1,086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	(2012)	82 min	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15501	Zulm Ki Hukumat	(1992)	0 min	Action, Crime, Drama	5.3	135	Bharat Rangachary	Dharmendra	Moushumi Chatterjee	Govinda
15503	Zulm Ki Zanjeer	(1989)	125 min	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15504	Zulm Ko Jala Doonga	(1988)	0 min	Action	4.6	11	Mahendra Shah	Naseeruddin Shah	Sumeet Saigal	Suparna Anand
15505	Zulmi	(1999)	129 min	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani

Next steps: New interactive sheet

```
movies['Year'] = movies['Year'].str.extract('(\\d{4})')
movies['Year'] = pd.to_numeric(movies['Year'])

movies['Duration'] = movies['Duration'].str.replace(' min', '')
movies['Duration'] = pd.to_numeric(movies['Duration'])

movies['Votes'] = movies['Votes'].astype(str)
movies['Votes'] = movies['Votes'].str.replace(',', '')
movies['Votes'] = pd.to_numeric(movies['Votes'])
```

```
/tmp/ipykernel_3610/3650544053.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Year'] = movies['Year'].str.extract('(\\d{4})')
/tmp/ipykernel_3610/3650544053.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Year'] = pd.to_numeric(movies['Year'])
/tmp/ipykernel_3610/3650544053.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Duration'] = movies['Duration'].str.replace(' min', '')
```

```
/tmp/ipykernel_3610/3650544053.py:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Duration'] = pd.to_numeric(movies['Duration'])
/tmp/ipykernel_3610/3650544053.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Votes'] = movies['Votes'].astype(str)
/tmp/ipykernel_3610/3650544053.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Votes'] = movies['Votes'].str.replace(',', '')
/tmp/ipykernel_3610/3650544053.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies['Votes'] = pd.to_numeric(movies['Votes'])
```

```
movies
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	2019	109	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	2019	110	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	1997	147	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	2005	142	Drama, Romance, War	7.4	1086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	2012	82	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15501	Zulm Ki Hukumat	1992	0	Action, Crime, Drama	5.3	135	Bharat Rangachary	Dharmendra	Moushumi Chatterjee	Govinda
15503	Zulm Ki Zanjeer	1989	125	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15504	Zulm Ko Jala Doonga	1988	0	Action	4.6	11	Mahendra Shah	Naseeruddin Shah	Sumeet Saigal	Suparna Anand
15505	Zulmi	1999	129	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani

Next steps: New interactive sheet

```
label_encoder = LabelEncoder()

categorical_columns = ['Genre', 'Director', 'Actor 1', 'Actor 2', 'Actor 3']
```

```
for col in categorical_columns:
    movies[col] = label_encoder.fit_transform(movies[col].astype(str))
```

```
/tmp/ipykernel_3610/4109197266.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies[col] = label_encoder.fit_transform(movies[col].astype(str))
/tmp/ipykernel_3610/4109197266.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies[col] = label_encoder.fit_transform(movies[col].astype(str))
/tmp/ipykernel_3610/4109197266.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies[col] = label_encoder.fit_transform(movies[col].astype(str))
/tmp/ipykernel_3610/4109197266.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies[col] = label_encoder.fit_transform(movies[col].astype(str))
/tmp/ipykernel_3610/4109197266.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-v
  movies[col] = label_encoder.fit_transform(movies[col].astype(str))
```

```
X = movies[['Year', 'Duration', 'Votes', 'Genre',
            'Director', 'Actor 1', 'Actor 2', 'Actor 3']]

y = movies['Rating']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
mae
```

```
0.8269482323232323
```

```
mse
```

```
1.2035198345959597
```

```
rmse
```

```
np.float64(1.0970505159726964)
```

```
r2
```

```
0.3526488017067818
```

This Movie Rating Prediction project demonstrates how machine learning can estimate movie ratings using features like genre, director, actors, duration, and votes. The project also highlights the importance of preprocessing and handling missing values before training a machine learning model.where r2 is not performing good and mse has more